

Mathematics A

Unit 1 • Chapter OL-T10 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 14 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T10 • Expertise Q16+ • 5 items • 14 marks

Chapter	Items	Marks	Starts
01. OL-T10 Surds	5	14	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Surds

Unit 1 • Expertise • 14 marks • 5 item(s)

June 2023 | 4MA1-2H Q23 | 4 marks

Binomial surd denominator using a conjugate

November 2024 | 4MA1-1H Q17 | 1 marks

Extract square factors from a surd

June 2022 | 4MA1-1H Q16 | 3 marks

Add or subtract like surds

January 2020 | 4MA1-1H Q17 | 3 marks

Expand binomials containing surds

May-Nov 2020 | 1H Q19 | 3 marks

Single-term surd denominator

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q23

4 marks

23 The diagram shows a cuboid with a square cross section.

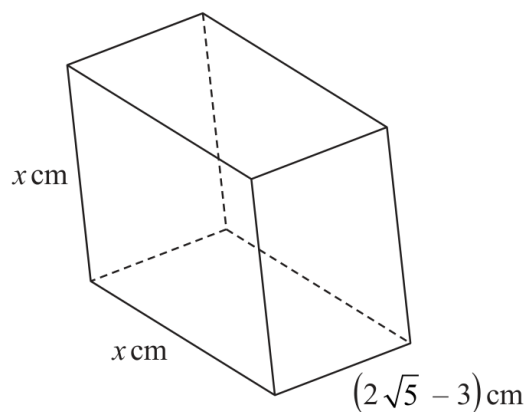


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x = \dots\dots\dots$

Continue your method**4 marks**

WORKING SPACE

4MA1-1H Q17 M01

1 marks

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

.....
(1)

WORKING SPACE

4MA1-1H Q16 Q16

3 marks

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$
Show your working clearly.

WORKING SPACE

17 (a) Show that $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(3)

YOUR WORKING

19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

(Total for Question 19 is 3 marks)

YOUR WORKING
